

World University of Bangladesh

Faculty of Science and Engineering

Department of Mechatronics Engineering

Program: Bachelor of Science in Automobile Engineering, Batch: AE 04B

Class Test: Spring 2026

Course Code: AM 07133109

Course Title: Material Testing & Fluid Power

Time: (30 +15) minutes

Full Marks: 15

Answer all of the following questions. Numbers shown in the right margin indicate full marks for the particular question or parts thereof.

- 1 (a) Write a short note on Brittleness and Ductility. 2
- (b) In a materials laboratory, a mild steel specimen is subjected to a tensile test using a Universal Testing Machine (UTM). The load is gradually increased and then removed at two different stages of the experiment. It is observed that when the load is removed at an initial stage, the specimen returns to its original length. However, when the load is removed after a higher level of stress is applied, the specimen does not regain its original dimensions and shows a permanent extension. 3
Based on this experimental observation and with the help of a stress-strain diagram, explain why the specimen behaves differently in the two cases.
- 2 (a) Define strength of a material and mention its main types with a brief explanation. 2
- (b) Differentiate between Destructive Testing (DT) and Non-Destructive Testing (NDT) of materials. Give one example of each type. 2
- (c) What is the purpose of Material Testing. 1
- 3 (a) Compare between Charpy Impact Test and Izod Impact Test. 2
- (b) A circular test bar specimen of diameter of $(80+X)$ mm and length 600 mm is subjected to elongation test. Test result shows: Gauge length $(200+X)$ mm, Yield Load 225 KN, Braking Load 242 KN, Final Length after elongation $(249+X)$ mm. Find the braking strength and percentage of elongation. [Here, X =Last Digit of your student ID] 3